Problem 6.9

At a certain rate of compound interest an investment of \$1,000 will grow to \$1,500 at the end of 12 years. Determine precisely when its value is exactly \$1,200.

Let the annual compound interest rate be i. Then

$$1500 = 1000(1+i)^{12}.$$

Dividing by 1000,

$$1.5 = (1+i)^{12}.$$

Taking the twelfth root,

$$1+i = (1.5)^{1/12}.$$

Now let t be the time in years when the investment is worth \$1,200. Then

$$1200 = 1000(1+i)^{t}.$$

Dividing by 1000,

$$1.2 = (1+i)^{t}.$$

Substituting $(1+i) = (1.5)^{1/12}$, we get

$$1.2 = \left((1.5)^{1/12}\right)^{t} = (1.5)^{t/12}.$$

Taking natural logarithms,

$$\ln(1.2) = \frac{t}{12}\ln(1.5).$$

So,

$$t = \frac{12\ln(1.2)}{\ln(1.5)}.$$

Therefore,

$$\boxed{t = \frac{12\ln(1.2)}{\ln(1.5)} \approx 5.40 \text{ years}}.$$